New area of circle $= \dfrac{\pi r^2}{4}$

$\therefore$ Decrease in area

$$= \pi r^2 - \frac{\pi r^2}{4} = \frac{3\pi r^2}{4}$$

Decrease percent in area

$$= \left(\frac{3}{4}\pi r^2 \times \frac{1}{\pi r^2} \times 100\right)\% = 75\%$$

Short trick Area is decreased by

$$= \left(-2x + \frac{x^2}{100}\right) = \left(-2 \times 50 + \frac{(50)^2}{100}\right)$$

$$= \left(-100 + \frac{2500}{100}\right) = -75\%$$

Area is decreased by 75%.

10. *(a)* Perimeter of square

$$= 4 \times 11 = 44 \text{ cm}$$

$2\pi r = 44$ cm [by condition]

$$\therefore \quad r = \frac{44}{2\pi} = 7 \text{ cm}$$

Thus, area of circle

$$= \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

11. *(b)* Circumference of circle

$$= 2\pi r = 2 \times \frac{22}{7} \times 42 = 264 \text{ cm}$$

$\therefore$ Length of wire = 264 cm

Wire is bent into a square.

$\therefore$ Perimeter of square = 264 cm

$\Rightarrow$ 4 × Sides of square = 264

$$\therefore \quad \text{Side of square} = \frac{264}{4} = 66 \text{ cm}$$

12. *(c)* Given, length of string = Radius of circle = 28 m

Area over which the horse can graze

$$= \pi r^2 = \frac{22}{7} \times 28 \times 28 = 2464 \text{ m}^2$$

13. *(c)* $\because$ Area of ring $= \pi(R^2 - r^2)$

Here, $r = 6$ cm, $R = ?$

$$\text{Area of ring} = 418 = \frac{22}{7}(R^2 - 6^2)$$

[given]

$$\Rightarrow \quad R^2 - 36 = \frac{418 \times 7}{22}$$

$$\Rightarrow \quad R^2 = 133 + 36 = 169$$

$$\Rightarrow \quad R = \sqrt{169} = 13 \text{ cm}$$

14. *(a)* Let the length and breadth of rectangle be l and b, respectively.

$\because$ Area of rectangle $= l \times b$ sq units

$$\therefore \quad \text{New length} = l + \frac{60}{100} \times l = \frac{8l}{5}$$

New breadth $= a$

$$\text{Then, } lb = \frac{8l}{5} \times a \quad \Rightarrow \quad a = \frac{5b}{8}$$

[by condition]

Decrease percent

$$= \left[\left(b - \frac{5}{8}b\right) \times \frac{1}{b} \times 100\right]\%$$

$$= 37\frac{1}{2}\%$$

15. *(b)* Let the length and breadth of rectangular field be $5x$ and $3x$, respectively.

$l = 5x$ and $b = 3x$

Perimeter of rectangular field $= 2(l + b)$

$$2(5x + 3x) = 240$$

$$\Rightarrow \quad 8x = 120 \Rightarrow x = 15$$

$$\therefore \quad l = 5x = 5 \times 15 = 75 \text{ m}$$

$$\text{and} \quad b = 3x = 3 \times 15 = 45 \text{ m}$$

Now, area of rectangle $= l \times b$

$$= 75 \times 45 = 3375 \text{ m}^2$$

16. *(a)* Given, inner circumference

$$= 2\pi r = 440 \text{ m}$$

$$\Rightarrow \quad r = \frac{440}{2 \times 22} \times 7 = 70 \text{ m}$$

Width of track = 14 m

$\because$ Radius of outer circle = (70 + 14) m = 84 m

$\therefore$ Diameter of outer circle

$$= 2 \times 84 = 168 \text{ m}$$

17. *(c)* Let length and breadth of rectangular be x and y, respectively.

The original area $= xy$

New length $= x + x \times 50\%$

$$= x + \frac{x}{2} = \frac{3x}{2}$$

New breadth $= y + y \times 20\%$

$$= y + \frac{y}{5} = \frac{6y}{5}$$

$$\text{New area} = \frac{3x}{2} \times \frac{6y}{5} = \frac{9}{5}xy$$

$$= \frac{9}{5} \text{ (original area)}$$

Short trick Area is increased by

$$= 50 + 20 + \frac{50 \times 20}{100} = 80\%$$

$$\text{New Area} = A + \frac{80}{100} \times A = \frac{9}{5}A$$

18. *(a)* Let each equal side $= a = 13$ cm and base $b = 24$ cm

$\therefore$ Area of the isosceles triangle

$$= \frac{1}{4} b \cdot \sqrt{4a^2 - b^2}$$

$$= \left[\frac{1}{4} \times 24 \times \sqrt{4 \times 169 - 24 \times 24}\right]$$

$$= 60 \text{ cm}^2$$

19. *(c)* Let the sides containing right angle be x cm and $(x - 14)$ cm.

Now, area of right triangle

$$= \left[\frac{1}{2} x \times (x - 14)\right] \text{cm}^2$$

But area = 120 cm^2 [given]

$$\Rightarrow \quad \frac{1}{2}x(x - 14) = 120$$

$$\Rightarrow \quad x^2 - 14x - 240 = 0$$

$$\Rightarrow \quad (x - 24)(x + 10) = 0$$

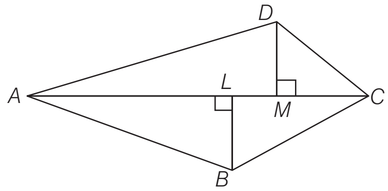
$$\Rightarrow \quad x \neq -10 \quad \therefore \quad x = 24$$

Other side = 24 − 14 = 10 cm

$$\text{Hypotenuse} = \sqrt{24^2 + 10^2} = \sqrt{676} \text{ cm} = 26 \text{ cm}$$

$\therefore$ Perimeter = (24 + 10 + 26) = 60 cm

20. *(d)* Here, $AC = 128$ m, $BL = 22.7$ m, $DM = 17.3$ m



$\therefore$ Area of the field

$$= \frac{1}{2}[AC(BL + DM)]$$

$$= \frac{1}{2} \times 128\,(22.7 + 17.3)$$

$$= 64 \times 40 = 2560 \text{ m}^2$$

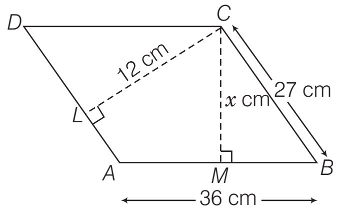
21. *(a)* Let distance between the longer sides be x cm.

Area of parallelogram

$$= AD \times LC = MC \times AB$$

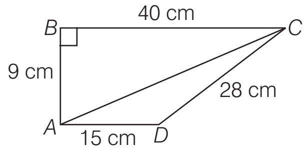
$$\therefore \quad 36 \times x = 27 \times 12$$

$$\Rightarrow \quad x = \frac{27 \times 12}{36} = 9$$



$\therefore$ Distance between the longer sides = 9 cm

22. *(b)* Applying pythagoras theorem in ΔABC, we get



$$9^2 + 40^2 = AC^2$$

$$\Rightarrow AC = \sqrt{1681} = 41 \text{ cm}$$

$$\text{Area of } \Delta ABC = \frac{1}{2} \times 9 \times 40 \text{ cm}^2$$

$$= 180 \text{ cm}^2$$

In ΔACD,

$$s = \frac{41 + 28 + 15}{2} = \frac{84}{2} = 42 \text{ cm}$$

$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$
$= \sqrt{42(42-41)(42-28)(42-15)}$
$= \sqrt{42 \times 1 \times 14 \times 27}$
$= 14 \times 3 \times 3 = 126 \text{ cm}^2$

$\therefore$ Area of quadrilateral = Area of ΔABC + Area of ΔADC
$= 180 + 126 = 306 \text{ cm}^2$

23. *(c)* Distance covered in one revolution
$= \dfrac{11 \times 1000 \times 100}{5000} = 220$ cm

$\therefore$ The circumference of the wheel $= 220$ cm

Let the diameter be 'D'.

Then, $\pi D = 220 \Rightarrow \dfrac{22}{7} \times D = 220$

$\therefore \quad D = \dfrac{220 \times 7}{22} = 70$ cm

24. *(a)* Let radius of given circles be x cm and $(14 - x)$ cm.

$\therefore$ Sum of areas of circle
$= [\pi x^2 + \pi(14-x)^2]$
$130\pi = \pi x^2 + \pi(14-x)^2$ [by condition]
$\Rightarrow \quad 130 = 2x^2 - 28x + 196$
$\Rightarrow \quad x^2 - 14x + 33 = 0$
$\Rightarrow \quad (x-11)(x-3) = 0$
$\Rightarrow \quad x = 11$ or $x = 3$

So, the radii of circles are 11 cm and 3 cm.

25. *(d)* Angle inscribed by minute hand in 60 min $= 360°$.

Angle inscribed in 35 min $= \dfrac{360°}{60} \times 35$
$= 210°$

given, $r = 12$ cm

$\therefore$ Area swept by the minute-hand in 35 min
= Area of sector with $r = 12$ cm and $\theta = 210°$

$= \dfrac{22}{7} \times \left(12 \times 12 \times \dfrac{210}{360}\right) \text{cm}^2 = 264 \text{ cm}^2$

26. *(b)* Area of the shaded region
= (Area of sector with $r = 7$ cm, $\theta = 30°$)
− (Area of sector with $r = 3.5$ cm, $\theta = 30°$)

$= \left[\left(\dfrac{22}{7} \times 7 \times 7 \times \dfrac{30}{360}\right) - \left(\dfrac{22}{7} \times \dfrac{7}{2} \times \dfrac{7}{2} \times \dfrac{30}{360}\right)\right] \text{cm}^2$

$= \left(\dfrac{77}{6} - \dfrac{77}{24}\right) \text{cm}^2 = \dfrac{77}{8} \text{ cm}^2$

27. *(d)* Area of field = 1 hec = 10000 m^2

$\therefore$ side $= \sqrt{10000}$ m = 100 m

$\because$ Side of other field = 102% of 100 = 102

$\therefore$ Area of the field
$= 102 \times 102 = 10404 \text{ m}^2$

Thus, difference of area
$= (10404 - 10000) \text{ m}^2$
$= 404 \text{ m}^2$

28. *(c)* Let diameter of the circle be $2r$.

$\therefore$ Area of the circle $= \pi r^2$

Diameter is increased by 100%

New diameter $= 4r$

and new area $= 4\pi r^2$

$\because$ Increase in area $= 4\pi r^2 - \pi r^2 = 3\pi r^2$

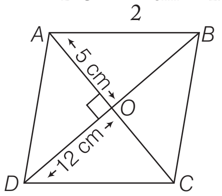
$\therefore$ Increase percentage in area
$= \left(\dfrac{3\pi r^2}{\pi r^2} \times 100\right)\% = 300\%$

Short trick Area is increased by
$= \left(2x + \dfrac{x^2}{100}\right)\% = \left(2 \times 100 + \dfrac{100^2}{100}\right)\%$
$= (200 + 100)\% = 300\%$

29. *(c)* Area of trapezium = 1/2 (Sum of parallel sides) × Distance between them
$= 1/2(25 + 15) \times 7 = 140 \text{ cm}^2$

30. *(b)* Since, diagonal of rhombus bisect each other and perpendicular to each other.

$\therefore \quad AO = \dfrac{10}{2}$ cm = 5 cm,
$DO = \dfrac{24}{2}$ cm = 12 cm



$\therefore \quad AD = \sqrt{OA^2 + OD^2} = \sqrt{5^2 + 12^2}$
$= \sqrt{169} = 13$ cm

$\therefore$ Perimeter $= 4 \times 13 = 52$ cm

31. *(a)* Here, $OA = OB = OC = 10$ cm

Also, $AC = 2CP$

and $CP = \sqrt{OC^2 - OP^2} = \sqrt{10^2 - 5^2}$
$= \sqrt{75} = 5\sqrt{3}$ cm

As, $AC = 2PC$

$\therefore \quad AC = 10\sqrt{3}$ cm

$\therefore$ Area of the rhombus $OABC$
$= \dfrac{1}{2} \times d_1 \times d_2 = \dfrac{1}{2} \times OB \times AC$
$= \dfrac{1}{2} \times 10 \times 10\sqrt{3} = 50\sqrt{3} \text{ cm}^2$

32. *(c)* Area of shaded region
= Area of ΔABP + Area of ΔPDC
$= \dfrac{1}{2} \times AB \times AP + \dfrac{1}{2} DC \times PD$
$= \dfrac{1}{2} \times AB \times AP + \dfrac{1}{2} AB \times PD$ $[\because AB = DC]$
$= \dfrac{1}{2} AB(AP + PD)$
$= \dfrac{1}{2} AB \times AD$ $[\because AD = (AP + PD)]$
$= \dfrac{1}{2} AB \times \dfrac{AB}{2} = \dfrac{1}{4} a^2$
$\left[\because AB = a \text{ and } AD = \dfrac{AB}{2}\right]$

33. *(a)* Area of figure = Area of rectangle + 2 (Area of semi-circular ends)
$= 20 \times 14 + 2 \times \dfrac{1}{2} \times \dfrac{22}{7} \times 7 \times 7$
$= 280 + 154 = 434 \text{ m}^2$

$\therefore$ Cost of levelling = ₹ (434×25)
= ₹ 10850

34. *(c)* Length of
$AC = AB + BC = 4 + 14 = 18$ cm

The area (shaded) bounded by three semi-circle = Area of semi-circle with AC as diameter − [(Area of semi-circle with diameter AB + Area of semi-circle with diameter BC)]

$= \dfrac{1}{2}\pi(9)^2 - \left[\dfrac{1}{2}\pi(7)^2 + \dfrac{1}{2}\pi(2)^2\right]$
$= \pi\dfrac{81}{2} - \dfrac{1}{2}[49\pi + 4\pi]$
$= \dfrac{\pi(81)}{2} - \dfrac{1}{2}[53\pi]$
$= \dfrac{\pi}{2}(81 - 53) = \dfrac{\pi(28)}{2} = 14\pi$

$\therefore$ Area of shaded region
= 14 times π

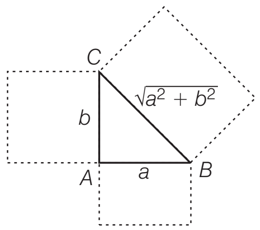
35. *(b)* Area of shaded region = Area of horizontal rectangle + Area of vertical rectangle
$= 5 \times 1 + (8-1) \times 1 = 5 + 7 = 12 \text{ m}^2$

36. *(b)* Length of boundary = Length of arc $(APB + BQC + ARC)$
$= \pi\left(\dfrac{AB}{2}\right) + \pi\left(\dfrac{BC}{2}\right) + \pi\left(\dfrac{AC}{2}\right)$
$= \pi\left(\dfrac{5}{2}\right) + \pi\left(\dfrac{5}{2}\right) + \pi(5)$
$= \dfrac{20\pi}{2} = 10\pi$ cm

37. *(d)* Radius of smaller circle = 2 cm

$\therefore$ Area of shaded region
$= \dfrac{1}{2}$[Area of larger circle − 2(Area of smaller circle)]
$= \dfrac{1}{2}[\pi 4^2 - 2(\pi)2^2]$
$= \dfrac{1}{2}[16\pi - 8\pi]$
$= \dfrac{1}{2}(8\pi) = 4\pi \text{ cm}^2$

38. *(c)*

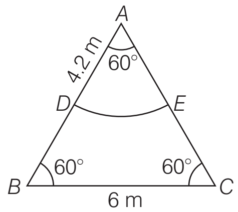


$\therefore$ Total area

$= a^2 + b^2 + (\sqrt{a^2 + b^2})^2 + \frac{1}{2}ab$

$= 2(a^2 + b^2) + 0.5ab$

39. *(c)* Suppose, a horse is tied at vertex A. Then, area available grazing field is ADE .



Now, area of sector $ADE = \frac{\pi r^2 \theta}{360°}$

$= \frac{22 \times (4.2)^2 \times 60°}{7 \times 360°} = 9.24 \text{ m}^2$

and area of equilateral

$\Delta ABC = \frac{\sqrt{3}}{4} (\text{Side})^2 = \frac{\sqrt{3}}{4} \times (6)^2$

$= 15.57$

$\therefore$ Required percentage $= \frac{9.24}{15.57} \times 100$

$= 59.34\% = 59\%$ (approx)

40. *(c)* Given, radius of circumcircle,

$r = 2$ units

side of square, $AB = 2$ units

Since, $EF = ME = MF = 2$ units

$\therefore MEF$ is an equilateral triangle

In ΔAEM, by pythagoras theorem

$AE^2 = EM^2 - AM^2 = (2)^2 - (1)^2$

$\therefore AE = \sqrt{3}$ units

Area of intersection region square and circle = Area of sector $MENF$ + Area of ΔAEM + Area of ΔFMB

$= \frac{\theta}{360°} \times \pi r^2 + \frac{1}{2} \times AE \times AM + \frac{1}{2} \times MB \times FB$

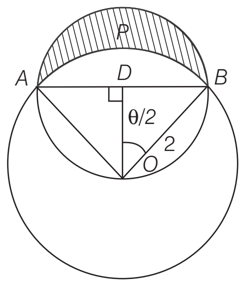
$= \frac{60°}{360°} \times \pi \times (2)^2 + \frac{1}{2} \times \sqrt{3} \times 1 + \frac{1}{2} \times \sqrt{3} \times 1$ $[\because FB = EA = \sqrt{3}]$

$= \frac{4\pi}{6} + \sqrt{3} = \frac{2}{3}\pi + \sqrt{3}$

41. *(b)* Given, radius, $r = 2$ cm

length of chord, $AB = 2\sqrt{2}$ cm

In ΔODB, $\sin\frac{\theta}{2} = \frac{BD}{OB}$



$\Rightarrow \sin\frac{\theta}{2} = \frac{\frac{2\sqrt{2}}{2}}{2} = \frac{1}{\sqrt{2}} \Rightarrow \frac{\theta}{2} = 45°$

$\therefore \theta = 90°$

Area of segment $APBA$ = Area of sector − Area of ΔOAB

$= \frac{1}{4}.\pi r^2 - \frac{1}{2} \times OA \times OB$

$= \frac{1}{4} \times \pi \times (2)^2 - \frac{1}{2} \times 2 \times 2$

$= (\pi - 2) \text{ cm}^2$

Area of shaded region = Area of semicircle with AB as diameter − Area of segment $APBA$

$= \frac{1}{2} \times \pi \times (BD)^2 - (\pi - 2)$

$= \frac{1}{2} \times \pi \times (\sqrt{2})^2 - (\pi - 2) = 2 \text{ cm}^2$

42. *(c)* We have, $a = l \times w$ ...(i)

and $p = 2(l + w)$ ... (ii)

Putting the value of l from Eq. (i) in (ii), we get

$wp - 2w^2 = 2a$

Now, $p^2 - 8a = [2(l + w)]^2 - 8lw$

$= 4(l^2 + w^2 + 2lw) - 8lw$

$= 4l^2 + 4w^2 + 8lw - 8lw$

$= 4(l^2 + w^2)$

Hence, the statement I and III are correct.

43. *(d)* I. Let R be the radius of circumcircle

Area of circumcircle $= \pi R^2$

$\Rightarrow 9\pi = \pi R^2 \Rightarrow R = 3$ cm

II. $\frac{\text{Inradius of polygon}}{\text{circumradius of polygon}} = \frac{r}{R}$

$= \frac{\frac{1}{2} a \cot\left(\frac{180°}{n}\right)}{\frac{1}{2} a \operatorname{cosec}\left(\frac{180°}{n}\right)} = \cos\left(\frac{180°}{n}\right)$

$\Rightarrow r = R\cos\left(\frac{180}{12}\right)^\circ = 3\cos 15°$

III. Area of polygon $= \frac{1}{2} nR^2 \sin\left(\frac{360°}{n}\right)$

$= \frac{1}{2} \times 12 \times R^2 \sin\left(\frac{360°}{12}\right)$

$= 6R^2 \sin 30° = 3R^2$

$\therefore \frac{\text{Area of circumcircle}}{\text{Area of polygon}} = \frac{\pi R^2}{3R^2} = \frac{\pi}{3}$

or $\pi : 3$

Hence, all three statements are correct.

44. *(c)* $\because$ From statement I

length = 2 × width

$\therefore$ Area of rectangle = 2 × width × width

= 2 (width)^2

From statement II,

$\therefore$ Area of rectangle = 2 × Perimeter

= 2 × 2(length + width)

= 4 (length + width)

From I and II,

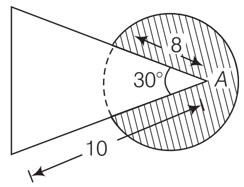
2 (width)^2 = 4 (2 width + width)

2 (width)^2 = 12 width

$\Rightarrow$ width = 6 units.

So, both statements together are sufficient.

45. *(d)* The length of the rope is 8 m, then the cow will be able to graze an area equal to the area of the circle with radius = 8 m, subtracting from that the area of the sector of the same circle with angle 30°,



which is equal to $\pi(8)^2 - \frac{\theta}{360}\pi(8)^2$

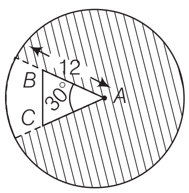
$= \pi \times 64 - \frac{30}{360}\pi \times 64 = \frac{176\pi}{3} \text{ m}^2$

46. *(c)* Since, the length of the rope is more than that of sides AB and AC,

$\therefore$ Required Area =

(area of the circle with radius 12) − (area of the sector of the same circle with angle 30°)

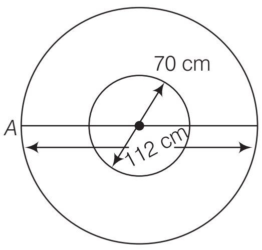
$= \pi(12)^2 - \frac{30}{360}\pi(12)^2 = 132\pi \text{ m}^2$



47. *(c)* $\because$ Outer diameter = 112 cm

and inner diameter = 70 cm

$\therefore$ Required area $= \frac{1}{4}\pi(112^2 - 70^2)$



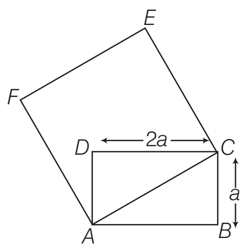
$= \frac{1}{4}(12544 - 4900)\pi$

$= \frac{1}{4} \times 7644 \times \frac{22}{7}$

$= \frac{1}{4} \times 24024 = 6006 \text{ cm}^2$

48. *(d)* Given that, Area of rectangle $= 2a^2$
$= l \times b$

$\Rightarrow \; l \times b = 2a^2 = l \times a \;\; \Rightarrow \;\; l = 2a$



Now, in ΔACD, by pythagoras theorem

$AC^2 = AD^2 + CD^2$

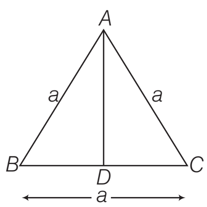
$= a^2 + 4a^2 = 5a^2$

$\therefore$ Side of square, $AC = a\sqrt{5}$ unit

Hence, area of the square $= (a\sqrt{5})^2$

$= 5a^2$ sq units

49. *(c)*



Height of equilateral triangle (AD)

$= \frac{\sqrt{3}}{2} \times \text{Side} \Rightarrow \sqrt{3} = \frac{\sqrt{3} \times \text{Side}}{2}$

$\Rightarrow 2\sqrt{3} = \sqrt{3} \times \text{Side}$

$\therefore$ Side $= \frac{2\sqrt{3}}{\sqrt{3}} = 2$ cm

$\therefore$ Perimeter of an equilateral triangle
$= 3a = 3 \times 2 = 6$ cm

50. *(a)* Let the width of the rectangle be x unit.

$\therefore$ Length $= (2x + 5)$ unit

According to the question,

Area $= x(2x+5) \Rightarrow 75 = 2x^2 + 5x$

$\Rightarrow \quad 2x^2 + 5x - 75 = 0$

$\Rightarrow \quad 2x^2 + 15x - 10x - 75 = 0$

$\Rightarrow \; x(2x+15) - 5(2x+15) = 0$

$\Rightarrow \quad (2x+15)(x-5) = 0$

$\therefore \quad x = 5$ and $\frac{-15}{2}$

Since, width cannot be negative.

$\therefore$ Width = 5 units

and length $= 2 \times 5 + 5 = 15$ units

$\therefore$ Perimeter of the rectangle
$= 2(15+5) = 40$ units

51. *(a)* Let the radius of circle is r and the side of a square is a, then by given condition,

$2\pi r = 4a \Rightarrow a = \frac{\pi r}{2}$

$\therefore$ Area of square $= \left(\frac{\pi r}{2}\right)^2 = \frac{\pi^2 r^2}{4}$

$= \frac{(3.14)^2 r^2}{4} = \frac{9.86r^2}{4} = 2.46r^2$

and area of circle $= \pi r^2 = 3.14r^2$

Again, let the side of equilateral triangle be x.

Then, by given condition, $3x = 2\pi r$

$\Rightarrow \; x = \frac{2\pi r}{3}$

$\therefore$ Area of equilateral triangle $= \frac{\sqrt{3}}{4}x^2$

$= \frac{\sqrt{3}}{4} \times \frac{4\pi^2 r^2}{9} = \frac{\pi^2}{3\sqrt{3}}r^2 = 1.89r^2$

Hence, Area of circle > Area of square
> Area of equilateral triangle

Hence, only I statements is correct.

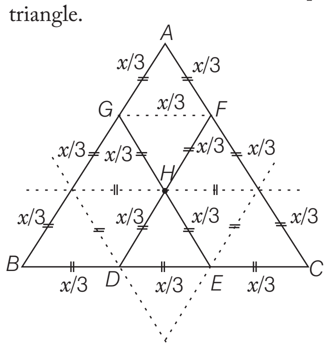
52. *(c)* Given that, Area of the circle = Area of the square = $(\text{Side})^2$

$\therefore \quad \pi r^2 = (2\sqrt{\pi})^2 \;\; \Rightarrow \;\; \pi r^2 = 4\pi$

$\Rightarrow \quad r^2 = \frac{4\pi}{\pi} = 4 \;\; \therefore \;\; r = \sqrt{4} = 2$ units

$\therefore$ Diameter of circle $(d) = 2 \cdot r = 2 \cdot 2$
$= 4$ units

53. *(b)* Here, ΔABC forms an equilateral triangle.



where, $AGHF$ form a rhombus and ΔHDE is also an equilateral triangle.

$\therefore$ Area of rhombus = (Area of ΔAGF
+ Area of ΔGFH)

$= \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2 + \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2 = 2 \cdot \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2$

Now, area of $\Delta HDE = \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2$

and area of $\Delta ABC = \frac{\sqrt{3}}{4}x^2$

By given condition,

$$\frac{\text{Area of rhombus } AGHF + \text{Area of } \Delta HDE}{\text{Area of } \Delta ABC}$$

$$= \frac{2 \cdot \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2 + \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2}{\frac{\sqrt{3}}{4}x^2}$$

$$= \frac{3 \cdot \frac{\sqrt{3}}{4}\left(\frac{x}{3}\right)^2}{\frac{\sqrt{3}}{4}x^2} = \frac{3}{9} = \frac{1}{3}$$

54. *(b)* We know that, the radius of a circle inscribed in a equilateral triangle $= a / 2\sqrt{3}$ where, a be the length of the side of an equilateral triangle.

Given that, area of a circle inscribed in an equilateral triangle = 154 cm^2

$\therefore \; \pi\left(\frac{a}{2\sqrt{3}}\right)^2 = 154$

$\Rightarrow \; \left(\frac{a}{2\sqrt{3}}\right)^2 = \frac{154 \times 7}{22} = 7 \times 7 = (7)^2$

$\Rightarrow \quad \frac{a}{2\sqrt{3}} = 7 \;\; \Rightarrow \;\; a = 14\sqrt{3}$ cm

$\therefore$ Perimeter of an equilateral triangle
$= 3a = 3(14\sqrt{3}) = 42\sqrt{3}$ cm

55. *(b)* Let the radii of two circles be r_1 and r_2, respectively.

Given, $\frac{\text{Circumference of Ist circle}}{\text{Circumference of IInd circle}} = \frac{2}{3}$

$\Rightarrow \frac{2\pi r_1}{2\pi r_2} = \frac{2}{3} \;\; \Rightarrow \;\; \frac{r_1}{r_2} = \frac{2}{3} \;\; \Rightarrow \left(\frac{r_1}{r_2}\right)^2 = \frac{4}{9}$ ...(i)

$\therefore \frac{\text{Area of Ist circle}}{\text{Area of IInd circle}} = \frac{\pi r_1^2}{\pi r_2^2} = \left(\frac{r_1}{r_2}\right)^2 = \frac{4}{9}$

or 4 : 9

56. *(c)* Let the radius of a circle be r and a be the length of the side of a square.

Given, circumference of a circle
= Perimeter of a square

$\Rightarrow \quad 2\pi r = 4a \;\; \Rightarrow \;\; a = \frac{\pi}{2}r = 1.57r$

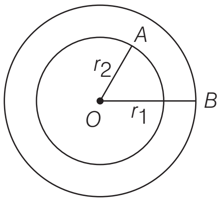
Now, area of the circle (A_c)
$= \pi r^2 = 3.14r^2$

and area of the square (A_s)
$= a^2 = 2.4649\,r^2$

$\therefore$ Area of circle > Area of square

57. *(c)* Given that, ratio of their radii = 5 : 3

i.e. $r_1 : r_2 = 5:3$



$$\Rightarrow \quad \frac{r_1}{r_2} = \frac{5}{3} \quad \text{...(i)}$$

Let, $r_1 = 5x$ and $r_2 = 3x$

Also, given that area enclosed between the circumferences of two concentric circles $= 16\pi \text{ cm}^2$

$\therefore \quad \pi(r_1^2 - r_2^2) = 16\pi$

$\Rightarrow (5x)^2 - (3x)^2 = 16$

$\Rightarrow \quad 25x^2 - 9x^2 = 16$

$\Rightarrow \quad 16x^2 = 16 \Rightarrow x^2 = 1$

$\therefore \quad x = 1$

$\therefore \quad r_1 = 5$ and $r_2 = 3$

$\therefore$ Area of the outer circle $= \pi r_1^2 = \pi(5)^2$
$= 25\pi \text{ cm}^2$

58. *(b)* Given that, perimeter of a rectangle
$= 82$ m

$\therefore$ 2 (Length + Breadth) = 82 m

$\Rightarrow$ Length + Breadth = 41 m

$\Rightarrow \quad l + b = 41 \text{ m} \quad \text{...(i)}$

Also, its area = 400 m^2

$\Rightarrow \quad l \cdot b = 400 \quad \text{...(ii)}$

Now, $(l - b)^2 = (l + b)^2 - 4lb$
$= (41)^2 - 4(400)$
$= 1681 - 1600 = 81$

$\therefore \quad l - b = 9 \quad \text{...(iii)}$

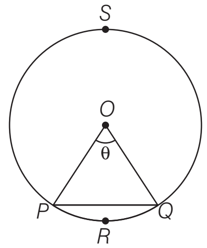
On solving Eqs. (i) and (iii), we get

$2l = 50 \Rightarrow l = 25$ m and $b = 16$ m

$\therefore$ Required breadth *(b)* = 16 m

59. *(c)* I. We know that, area of segment (*PRQP*)
= Area of sector (*OPRQO*)
– Area of ΔOPQ

$$= \frac{\pi r^2 \theta}{360} - \frac{1}{2} r^2 \sin\theta$$



So, the area of a segment of a circle is always less than area of its corresponding sector.

II. Distance travelled by a circular wheel of diameter $2d$ cm in one revolution

$$= 2\pi \frac{(2d)}{2} = 2 \times 3.14 \times d = 6.28\, d$$

which is greater than $6d$ cm.

Hence, both statements are correct.

60. *(a)* $\because$ Angle described in 60 min by minute hand of a clock $= 360°$

and angle described in 40 min by minute hand of a clock

$$= \frac{360°}{60°} \times 40 = 240°$$

$$\therefore \text{Required distance} = \frac{2\pi(2 \cdot 5) \times 240°}{360°} = \frac{10\pi}{3} \text{ cm}$$

61. *(d)* Given that, length of hour hand
$= 4$ cm

and length of minute hand = 6 cm

$\therefore$ Hour hand rotating in 1 day
$= 2 \times 360° = 720°$

$\therefore$ Hour hand rotating in 2 days
$= 2 \times 720° = 1440°$

Similarly,

Minute hand rotating in 1 day
$= 24 \times 360°$

$\therefore$ Minute hand rotating in 3 days
$= 72 \times 360°$

$\therefore$ Distance travelled by hour hand

$$= \frac{2\pi(4) \times 1440°}{360°} = 32\pi$$

and distance travelled by minute hand

$$= \frac{2\pi(6) \times 72 \times 360°}{360}$$

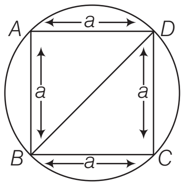
$= 6 \times 144\pi$ cm

$\therefore$ Required ratio $= 32\pi : 6 \times 144\pi$
$= 1 : 27$

62. *(c)* Let side of a square be a cm.

Given that, radius of a circle = 8 cm

and diameter of a circle = 16 cm



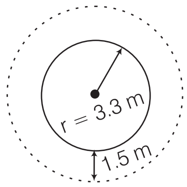
$\because$ Length of a diagonal of a square
= Diameter of a circle

$\Rightarrow \quad a\sqrt{2} = 16$

$\therefore \quad a = 8\sqrt{2}$ cm

$\therefore$ Area of square $ABCD = a^2 = (8\sqrt{2})^2$
$= 64 \times 2 = 128 \text{ cm}^2$

63. *(c)* Area of the path = Area of circular water [fountain + path]
– Area of circular water fountain



$= \pi(3.3 + 1.5)^2 - \pi(3.3)^2$

$= [(4.8)^2 - (3.3)^2]\,\pi$

$= (23.04 - 10.89)\,\pi$

$= 12.15\pi \text{ m}^2$

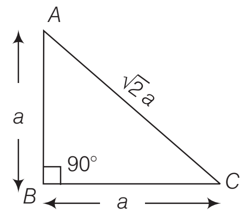
64. *(d)* Let the equal sides of isosceles right triangle be a cm.

Then $AB = BC = a$

In ΔABC, by pythagoras theorem

$AC^2 = AB^2 + BC^2 = 2AB^2$

$\Rightarrow \quad AC^2 = 2a^2 \quad \therefore \quad AC = \sqrt{2}a$



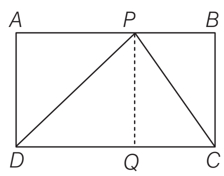
$$\text{Area of } \Delta ABC = \frac{1}{2} \times a \times a$$

$$\Rightarrow \quad 1 = \frac{1}{2}a^2 \Rightarrow a = \sqrt{2}$$

$\therefore$ Perimeter of $\Delta ABC = 2a + a\sqrt{2}$
$= 2\sqrt{2} + \sqrt{2} \cdot \sqrt{2}$
$= 2(\sqrt{2} + 1)$ units

65. *(c)* Given that, $CD = 20$ cm

and area of rectangle $ABCD = 100 \text{ cm}^2$



$\Rightarrow AD \times CD = 100 \text{ cm}^2$

$\Rightarrow \quad AD \times 20 = 100$

$\therefore \quad AD = 5$ cm

$$\therefore \quad \text{Area of } \Delta PDC = \frac{1}{2} \times PQ \times CD$$

$$= \frac{1}{2} \times 5 \times 20 = 5 \times 10 = 50 \text{ cm}^2$$

$[\because AD = PQ]$

66. *(c)* Area of parallelogram
= Base × Height
$= 8.06 \times 2.08 = 16.76 \text{ cm}^2$